

Week 4 — Research-Level Full System

Goal: Integrate everything into a polished research-grade pipeline.

Day	Key Tasks
Day 1	Gradio UI setup, Webcam dashboard
Day 2	Adaptive resolution controller, CPU fallback tiers
Day 3	Evaluation framework, BLEU/ROUGE metrics
Day 4	Dataset collection, Ground-truth captions
Day 5	Ablation study A–E, Metric comparison tables
Day 6	Research paper drafting, Results graphs
Day 7	GitHub polish, Final demo preparation

Expected Deliverables

- Complete Gradio application
- Research metrics collected
- Ablation studies completed
- Paper-ready system

This week contributes directly toward building a complete real-time scene understanding research system capable of object detection, depth estimation, segmentation, narration, and intelligent perception.